Your grade: 96%

Your latest: **90%** • Your highest: **96%** • To pass you need at least 80%. We keep your highest score.

Next item

coach

Almost there! Let's review your answers and then try again.

Your work showed a good understanding of tensors, backward passes, and the role of the `torch.save` function in PyTorch. However, there is room for improvement in understanding the PyTorch Dataloader class. Before retrying this assignment, we recommend focusing on this key area:

- **Datasets and Dataloaders (101)**: In Question 3, your understanding of the actions that can be performed using the PyTorch Dataloader class was incomplete. Try reviewing this item to refresh your memory.

Good job.

Retry assignment

1. Which of the following are properties of tensors in PyTorch? — 1 / 1 point

- [ ] Inherent data visualization
- [x] Dependency tracking

Correct
Correct! Dependency tracking is a key feature of tensors in PyTorch, allowing for efficient backpropagation.

- [ ] Built-in data normalization
- [x] Ability to run on GPU

Correct
Correct! Tensors in PyTorch can be transferred to and run on a GPU for faster computation.

- [x] Automatic gradient calculation

Correct
Correct! Tensors in PyTorch have automatic gradient calculation capabilities that are essential for training neural networks.

2. Which function is used to perform a backward pass in PyTorch? — 1 / 1 point

- ○ autograd()
- ○ forward()
- ● backward()
- ○ tensor()

Correct
Correct! The backward() function in PyTorch is used to perform a backward pass to compute the gradient of a tensor.

3. Which of the following actions can be performed using the PyTorch Dataloader class? — 0.5 / 1 point

- [ ] Automatically splitting the data into training and testing sets
- [ ] Loading data in parallel using multiprocessing
- [x] Shuffling the data

Correct
Correct! PyTorch Dataloader can shuffle the data to ensure that the model does not learn the order of the data.

- [ ] Sampling the data based on specified probabilities
- [x] Collating data into batches

Correct
Correct! PyTorch Dataloader collates data into batches for efficient model training.

- [x] Normalizing the data

This should not be selected
Incorrect. Normalizing the data is typically done using torchvision.transforms or custom transformations, not directly within the Dataloader class.

4. What is the role of the torch.save function in PyTorch? — 1 / 1 point

- ● To save the state dictionary of a model for future use
- ○ To define the model architecture
- ○ To train the model
- ○ To initialize model parameters

Correct
Correct! The torch.save function is used to save the state dictionary of a model so it can be loaded and used later.

5. What is the primary purpose of partitioning data into batches during model training? — 1 / 1 point

- ○ To ensure different data distributions are used in training
- ● To efficiently use computational resources and stabilize gradient updates
- ○ To simplify the implementation of the model
- ○ To increase the complexity of the model

Correct
Correct! Partitioning data into batches helps efficiently use computational resources and stabilizes gradient updates.